PRACTICAL-6

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Cost Estimation using COCOMO technique:

Code:

```
import math

print("COCOMO COST ESTIMATION")
print("1. Organic")
print("2. Semi-Detached")
print("3. Embedded")

mode = int(input("Enter mode (1-3): "))
kloc = float(input("Enter project size in KLOC: "))
cost_per_month = float(input("Enter cost per person-month (Rs.): "))

if mode == 1:
  a, b, c, d = 2.4, 1.05, 2.5, 0.38
  mode_name = "Organic"

elif mode == 2:
  a, b, c, d = 3.0, 1.12, 2.5, 0.35
  mode_name = "Semi-Detached"

elif mode == 3:
  a, b, c, d = 3.6, 1.20, 2.5, 0.32
```

```
    mode_name = "Embedded"

else:
    print("Invalid mode")
    exit()

effort = a * math.pow(kloc, b)
development_time = c * math.pow(effort, d)
staff = effort / development_time
total_cost = effort * cost_per_month

print("\n----- COCOMO COST ESTIMATION -----")
print("Mode :", mode_name)
print("Project Size :", kloc, "KLOC")
print("Effort :", round(effort, 2), "Person-Months")
print("Development Time :", round(development_time, 2), "Months")
print("Staff Required :", round(staff, 2), "Persons")
print("Estimated Cost : Rs.", round(total_cost, 2))
```

Outputs:

1. Organic:

Output

```
COCOMO COST ESTIMATION
1. Organic
2. Semi-Detached
3. Embedded
Enter mode (1-3): 1
Enter project size in KLOC: 10
Enter cost per person-month (Rs.): 50000

----- COCOMO COST ESTIMATION -----
Mode : Organic
Project Size : 10.0 KLOC
Effort : 26.93 Person-Months
Development Time : 8.74 Months
Staff Required : 3.08 Persons
Estimated Cost : Rs. 1346422.15
```

2. Semi-detached:

Output

```
COCOMO COST ESTIMATION
1. Organic
2. Semi-Detached
3. Embedded
Enter mode (1-3): 2
Enter project size in KLOC: 20
Enter cost per person-month (Rs.): 60000

----- COCOMO COST ESTIMATION -----
Mode : Semi-Detached
Project Size : 20.0 KLOC
Effort : 85.96 Person-Months
Development Time : 11.88 Months
Staff Required : 7.23 Persons
Estimated Cost : Rs. 5157344.0
```

3. Embedded:

Output

```
COCOMO COST ESTIMATION
1. Organic
2. Semi-Detached
3. Embedded
Enter mode (1-3): 3
Enter project size in KLOC: 30
Enter cost per person-month (Rs.): 75000

----- COCOMO COST ESTIMATION -----
Mode : Embedded
Project Size : 30.0 KLOC
Effort : 213.23 Person-Months
Development Time : 13.91 Months
Staff Required : 15.33 Persons
Estimated Cost : Rs. 15992238.94
```